

Sri Lanka Institute of Information Technology



IT1040 - Fundamentals of Computing
Year 1, Semester 1- 2024

ClothMate - A Home-autonomous system

Progress Report
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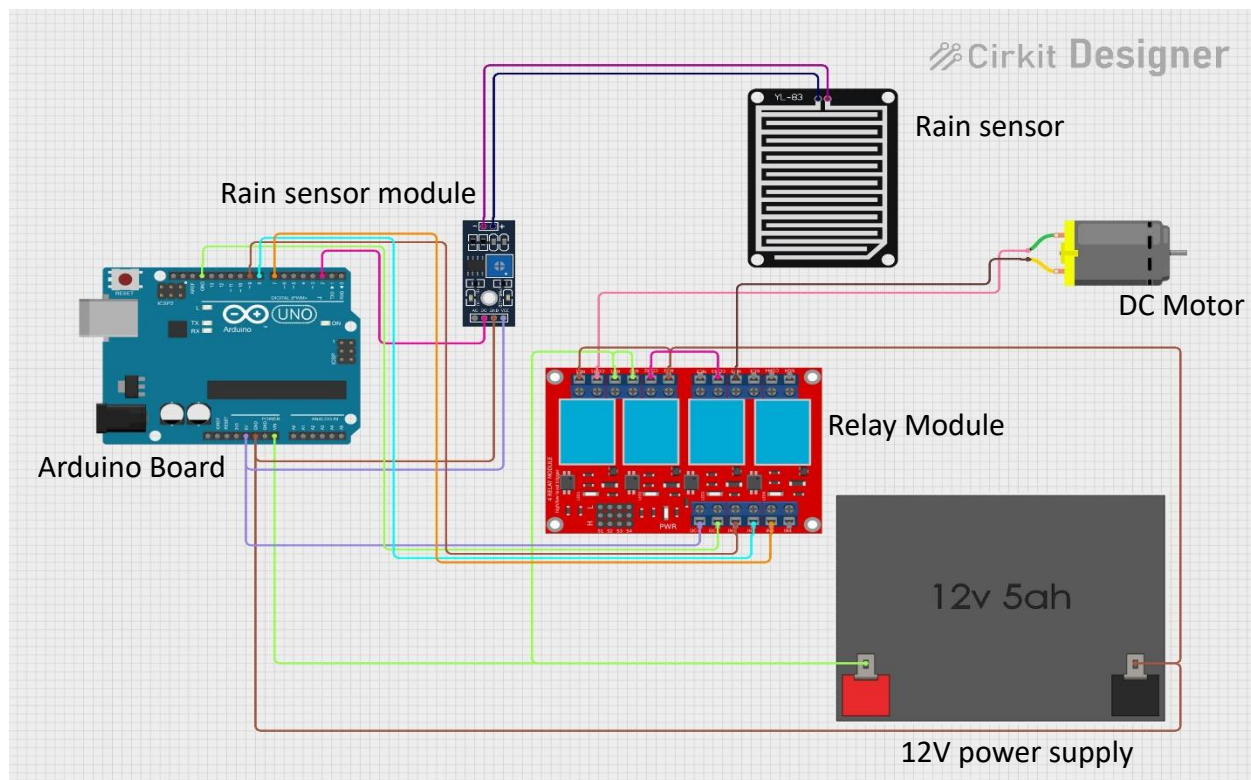
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1. Introduction

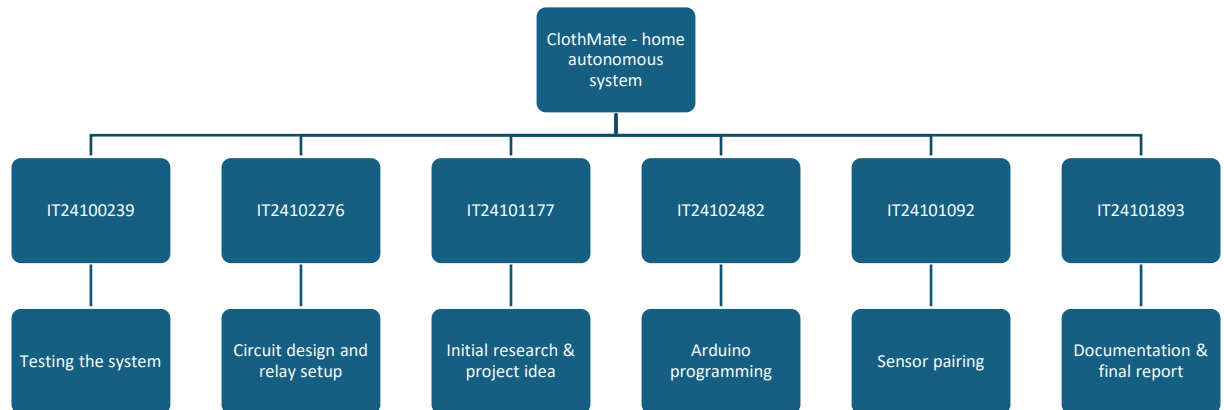
Our project concerns developing an automated clothesline system called Clothmate that seeks to improve the ease obtained when drying clothes. Clothmate has a weather sensing ability which allows it to retract clotheslines for the dry clothes when it starts raining. The project utilizes the Arduino board for automation which makes the project affordable and easier.

2. System Architecture

1. **Sensors:** Detect environmental conditions such as rain.
2. **Arduino Board:** Acts as the central controller, processing input from sensors and activating the relays accordingly.
3. **Relays:** Control the motor responsible for retracting and extending the clothesline. Three relays are used:
 - **Relay 1:** Controls the extension of the clothesline.
 - **Relay 2:** Controls the retraction of the clothesline.
 - **Relay 3:** Can be used for an emergency stop or override functionality.
4. **Motor:** Activated by the relays to physically move the clothesline.
5. **Power Supply:** Powers the Arduino and the relays.



3. Work Breakdown Structure



4. Proof of Work

